

Yining Wen

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EDUCATION

The University of Alabama

Ph.D. in Civil, Construction and Environmental Engineering

Master of Science in Electrical and Computer Engineering

New York University

Master of Science in Mechanical Engineering

Beihang University

Bachelor of Engineering in Aeronautic Science and Engineering

Tuscaloosa, AL

Expected Dec. 2026

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New York, NY

May 2020

Beijing, China

May 2018

SKILLS

Programming: Python, C++, MATLAB

3D Vision: DINO, NeRF, 3DGS, DeepLab, YOLO, Capsule, SAM

ML/AI: PyTorch, ONNX, TensorRT, Transformers, VLM, LLM

Robotics & Spatial: ROS, Omniverse, SLAM, NBV, MIMO

RESEARCH PROJECTS

3D Spatial Reasoning for Physical Simulation

Dec. 2025 ~ Aug. 2026

- Developed a multimodal **VLM** pipeline that maps RGB/IR imagery into 3D reconstructed representation, and extracts spatial information (e.g., geometry, parent relationships, etc.) for downstream physical simulation.
- Developed multimodal vision encoders (**DINOv3+DPT+SAM2, Capsule**) to extract object-level 2D features and aligned semantic segmentations into a **Gaussian Splatting**-based 3D representation to recover spatially grounded geometry.
- Integrated 3D object-level spatial tokens into a **QLoRA**-fine-tuned **Mistral-7B** to generate physically consistent inputs for downstream energy simulation.
- Applied **RLHF** boosted structure validity from 70% to 85%, spatial consistency from 58% to 75% compared to SFT-only training, and reduced schema violation by 34%.

Edge Autonomy for UAV-to-Digital Twin Inspection (NSF Project ID: 2531557)

Aug. 2025 ~ Present

- Architected a UAV-DT inspection system integrating **Pixhawk**, **NVIDIA Jetson Orin Nano**, and **NVIDIA Omniverse**, enabling building inspectors to remotely conduct autonomous inspections without piloting expertise.
- Developed onboard **YOLOv8** (~98% mIoU) for RGB defects and **Capsule** (~83% mIoU) for infrared defects detection via **PyTorch, ONNX, TensorRT**.
- Developed a **Transformer-based, uncertainty-driven Next-Best-View (NBV) planner** to reposition the UAV to acquire higher-fidelity viewpoints, improving real-time detection performance by 4.9%~22.8%.
- Built a UAV-DT pipeline in **NVIDIA Omniverse**, synchronizing 3D environment with defects and updated viewpoints, giving inspectors a clearer, higher-confidence view for **physical-virtual interaction** and downstream analysis.

Thermal Perception with Human-AI Active Learning

Aug. 2023 ~ Aug. 2024

- Developed a **Human-in-the-Loop Active Learning** system for thermal anomaly segmentation, extracting physically interpretable defect cues (e.g., heat leaks, insulation voids) with scarce labels.
- Designed **uncertainty-driven sampling** and human-AI collaborative annotation to prioritize high-value frames; Integrated outputs as the perception layer for downstream digital-twin reasoning and Jetson onboard inspection modules.
- Reduced segmentation annotation cost by 40%-59%; cut expert labeling time by 26%; improved detection performance from 0.32 to 0.65 under limited annotation budgets.

Automotive Radar Perception -- NXP Semiconductor

Aug. 2021 ~ Jan. 2023

- Developed radar perception algorithms for autonomous driving combining sparse signal reconstruction and deep learning to improve object localization and environment perception. synthesize a 76λ-aperture virtual array from only 14 physical antennas (16 Tx + 8 Rx), recovering ~68% missing elements to reach 0.67° angular resolution.
- Implemented accelerated **Forward-Backward Iterative Hard Thresholding (FB-IHT)** for Hankel matrix completion to lower the beam-vector recovery error and raise detection success rate across 0-30 dB SNR and up to 9 simultaneous targets, while cutting per-iteration compute by avoiding full SVD.
- Designed an **unrolled deep neural network** for radar signal matrix completion, integrating model-based optimization with learnable architectures to improve reconstruction accuracy while reducing inference latency.

LINKS

Google Scholar: <https://scholar.google.com/citations?user=SsvykR0AAAAJ&hl=en>